

Joe Chen

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Availability: April – December 2026

EDUCATION

Northeastern University, Khoury College of Computer Sciences

Boston, MA

Bachelor of Science in Computer Science, Minor in Mathematics

GPA: 3.73/4.0 | Sept 2024 - May 2028

Relevant Coursework: Algorithms and Data, Computer Systems, Machine Learning and Data Mining 1, Machine Learning and Data Mining 2, Practical Neural Networks, Probability and Statistics, Linear Algebra

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, C, HTML/CSS, SQL, R

Frameworks & Libraries: PyTorch, TensorFlow, React, Node.js, Next.js, UNIX/Linux, Prisma, OpenCV

Tools: Git/GitHub, Jupyter, Figma, Neon, npm

PROJECTS

AI Local Assistant | *Python, Pytorch, OpenCV, Mediapipe, Faster-Whisperer*

Sept 2025 - Present

- Prototyped a fully local AI assistant using 4-bit quantized Llama-3.2-11B-Vision-Instruct, achieving >50% VRAM reduction while preserving multimodal reasoning capabilities with zero cloud dependency
- Integrated speech-to-text (Faster-Whisper), computer vision (OpenCV + MediaPipe), and a custom Siamese CNN for face recognition (98.8% accuracy), enabling real-time multimodal interaction through a modular, thread-safe pipeline
- Designed a layered perception architecture using MediaPipe for real-time face authentication and gesture tracking, with modular interfaces built to support future ML classifier integration

Resona - Music Analytics App | *TypeScript, Next.js, PostgreSQL, Prisma, Python*

Sept 2025 - Present

- Developed a Next.js full-stack app where users rate tracks, albums, and artists (0–10), auto-generating social feed posts with threaded comments, likes, and follower graphs via a custom Prisma ORM schema over NeonDB
- Integrated Google OAuth via NextAuth v5 with callback-level waitlist gating, Spotify OAuth token storage, Vercel Blob for image uploads, and optimistic client-side state updates
- Building a FastAPI microservice for collaborative filtering recommendations using Funk SVD matrix factorization and cosine similarity on latent user-rating vectors

EXPERIENCE

Visual Intelligence Lab @ Northeastern University

Boston, MA

Undergraduate Research Assistant

March 2026 - Present

- Built a synthetic relighting data pipeline on a GPU cluster using Infinigen and a custom Blender Python script to re-render 3D scenes under varied lighting, generating paired training data for LumiTokens — a novel token-space relighting model (ECCV 2026 submission)
- Configured the Infinigen pipeline on SLURM, resolving undocumented dependency conflicts and optimizing synthetic scene rendering across A5000/A6000 GPUs at scale

BCNC Red Oak After School & Summer Program

Boston, MA

Group Leader

Sept 2023 - Aug 2024

- Supervised ~20 children daily, fostering a safe, engaging environment across academic and recreational activities
- Organized and facilitated 10+ enrichment events that enhanced student engagement, curiosity, and problem-solving skills

LEAH STEM Teaching Internship

Boston, MA

Youth Leader

Oct 2022 - May 2023

- Delivered 15+ hours of interactive STEM instruction by developing and executing 10+ hands-on lessons in biology, chemistry, and physics, cultivating curiosity and problem solving with ~15 elementary students
- Mentored students consistently across 8 months, serving as a role model and consistent mentor